

Impairments of Learned Aversion Acquisition Following Paradoxical Sleep Deprivation in the Rat

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DANGUIR, J. AND S. NICOLAIDIS. *Impairments of learned aversion acquisition following paradoxical sleep deprivation in the rat.* PHYSIOL. BEHAV. 17(3) 489–492, 1976. — The effect of paradoxical sleep deprivation (PSD) on taste-aversion learning was studied utilizing the classical water tank technique. When such a PSD was applied for 48 hr preceding and 48 hr following the learning session (LS), taste-aversion learning was completely abolished. A slightly attenuated effect was obtained when the entire 96 hr of PSD preceded the LS. When the 96 hr PSD followed the LS, learning was only slightly affected. The effect of PSD was duration dependant, starting at 7.5 hr pre-LS followed or not by the same duration of PSD post-LS. We conclude that this technique, thought to induce almost complete PSD in addition to nonspecific stress, affects particularly the acquisition processes (pre-LS PSD) and only slightly those of retention (post-LS PSD).

Taste-aversion learning
Acquisition processes

Paradoxical sleep deprivation
Retention processes

Water tank technique

Lithium chloride poisoning

PARADOXICAL sleep (PS) seems to have a key role in the processes of classical learning and conditioning. Ample evidence for this assertion has been gathered using passive avoidance techniques [4, 16, 20], or active avoidance paradigms [1, 7, 9, 20, 23].

Recently, there has been considerable interest in what appears to be a highly specialized form of learning, namely that of long-delay associations studied principally by Garcia, Rozin, Nachman and their coworkers [5, 6, 14, 18]. It has been shown that only one exposure to an olfacto-gustatory stimulus (conditional stimulus, CS) is sufficient to form a strong association with sickness (unconditional stimulus, US) induced by lithium chloride, apomorphine or X-rays. This association persists even when the interval between CS and US is as much as 12 hr. The phenomenon is more robust if the CS is a novel and salient substance [17, 22]. This learning is thus distinct from other kinds of acquired processes not only because of the possible long delay between CS and US, but also because the aversion is strong and persistent.

Since the neurophysiological mechanisms seem to be different from those of classical learning and conditioning processes, we have tried to elucidate the role of PS in the long delay learning. In a first experiment, we established a correlation between the manifestation of a learned taste aversion and the duration of paradoxical sleep deprivation (PSD) preceding and following the learning session (LS). We then investigated the possibility that acquisition and retention may be different processes which are differentially

affected by PSD, and so deprived animals of PS either before or after the LS.

GENERAL METHOD

Animals

One hundred and twelve male rats of the Wistar strain, weighing 200–250 g at the beginning of the experiment, were used.

Apparatus

Paradoxical sleep deprivation was produced by the classical water tank technique [12]. In this technique, rats were required to maintain their balance on a small (diameter 7 cm) pedestal just above the surface of water, which was shallow enough to allow the rats to climb back on the pedestal if they should fall off. This procedure is claimed to deprive the rat of at least 80–100% PS [21], while allowing slow wave sleep to occur (only slightly attenuated).

Control rats were put in similar conditions except that the diameter of the pedestal in the water tank was 20 cm. Pedestals of a diameter greater than 11.5 cm allow the rat to develop complete sleep cycles [21]. When not in the water tank where lighting was continuous, all rats were kept in individual wire mesh cages in a room illuminated from 08.00–20.00 hr.

The animals on pedestals had daily breaks of 75 min for

feeding in their home cages. After feeding, they were allowed a further 15 min for drinking water, saline, or lithium chloride, depending on the experiment. During this time, the rats were kept under observation and prevented from falling asleep if necessary.

EXPERIMENT 1

In this first study, we examined the effect of PSD upon conditioned taste-aversion. Two groups of rats (D_{96} and C_3) were used in this experiment. As schematized on Fig. 1, at 10.00 on Day 1 both groups were placed in the water tank. On Day 2, all rats were given water. On Day 3, D_{96} rats received 0.15 M LiCl solution to drink while C_3 control rats were given an amount of isotonic saline equal to the average consumed by D_{96} rats. On Day 4, all rats of both groups (D_{96} and C_3) were offered water. On Day 5 through Day 9 all rats were returned to the home cages with food and water available ad lib. On Day 9, all rats were deprived of water for 24 hr and on Day 10 they were presented a 0.9% NaCl solution and the amount of drinking (± 0.5 ml) during 15 min period was measured. It has been shown that rats are unable to distinguish between the salty tastes of NaCl and LiCl [13]. Thus, when they have been poisoned with water containing LiCl, they avoid both LiCl and NaCl in the future.

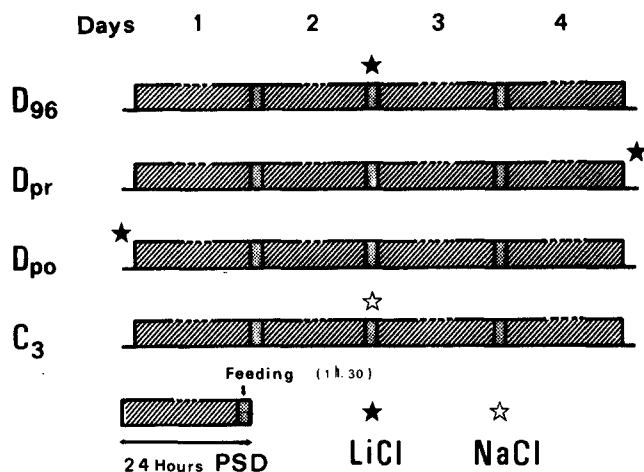


FIG. 1. Schematic representation of paradoxical sleep deprivation (hatched bars) interrupted by daily breaks (stippled bars) of feeding and drinking water, NaCl (*) or LiCl (*) for each group deprived from paradoxical only before (D_{pr}), only after (D_{po}) or around (D_{96}) the learning session. C_3 = control group.

Two other control groups (C_1 and C_2) received the same treatment as D_{96} rats except that there was no interference with their PS, since Group C_2 lived on 20 cm pedestals and Group C_1 were kept in their home cages.

RESULTS

Both D_{96} rats ($N = 8$) and C_3 rats ($N = 6$) drank on Day 10, about 15 ml of NaCl test solution (15.0 ± 1.4 ml and 15.2 ± 1.2 ml respectively). These intakes did not differ significantly ($p > 0.1$; Student t test). However, rats of control Group C_2 , and of course rats of control Group C_1 , showed a strong aversion to the salty taste ($p < 0.001$) (Fig. 2).

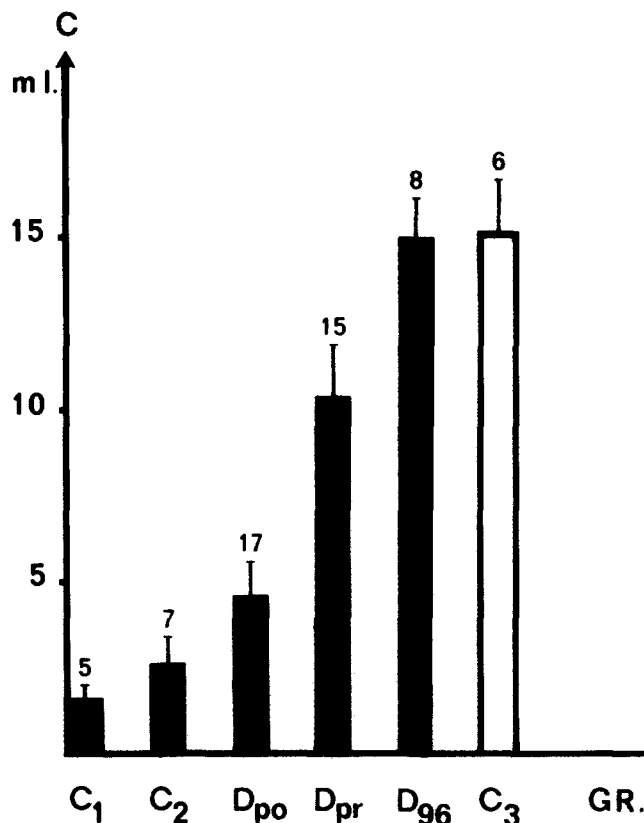


FIG. 2. Mean consumption and standard error (topped by N) of saline made aversive (except for C_3), by LiCl poisoning in control groups (C_1 and C_2) and in PSD rats before (D_{pr}), after (D_{po}) or before and after (D_{96}) the learning session.

It is evident that a long period of PSD preceding and following the LS led to no behavioral evidence for learning and retention of a taste aversion.

EXPERIMENT 2

In order to find out what is the minimal necessary PSD preceding and/or following the LS for interference with the taste aversion learning, a new series of 4 groups was tested. The durations of PSD were 40 hr (Group D_{40}), 20 hr (Group D_{20}), 15 hr (Group D_{15}) and 10 hr (Group D_{10}) respectively. The same protocol was applied as in Experiment 1: the rats were subjected to the water tank conditions for half of the total PSD period. They were then allowed to drink LiCl solution and were again put in the water tank for the second half of the PSD period. After recovery phase in their home cages, the rats were offered NaCl test solution for 15 min. It is important to note that the major part of the periods of PSD occurred during the day since schedules of deprivation were calculated before and after the LS at 10.00 a.m. (i.e. 2 hr into the day time).

RESULTS

The NaCl intake of the different groups is shown in Fig. 3. Ten hours of PSD (5 hr before the LS and 5 hr after it) did not significantly affect the avoidance of salt solutions. PSD of 15 hr (7.5 hr before the LS and 7.5 hr after it) significantly affected this avoidance ($p < 0.01$).

EXPERIMENT 3

In the two previous experiments rats were deprived of PS both before and after the LS. The observed decreases in aversion learning could be attributed to either the processes of acquisition or of retention. It seems reasonable to attribute any effect of PSD preceding the LS to an alteration of acquisition mechanisms, while PSD following the LS would most likely affect the retention processes of the acquired avoidance. In order to discriminate between the roles of acquisition and retention in taste-aversion learning, we applied PSDs only preceding or only following the LS.

RESULTS

In Fig. 2, D_{pr} represents the degree of impairment of the aversion to salt taste when PSD was restricted to a period of 96 hr preceding the LS. This moderate reduction of learned avoidance is in contrast to that of rats which were deprived of PS only after the LS (Group D_{po}). Both values significantly differ from D_{96} rats. Interestingly, the sum of the amounts consumed by Group D_{pr} and Group D_{po} is practically equal to the amount taken by Group D_{96} as if there were an additivity of effects due to PSDs preceding and following the LS.

EXPERIMENT 4

Experiment 3 showed that acquisition processes might be more affected than those of retention. We therefore tried to find out the quantitative relationship between the magnitude of PSD and the reduction of the learned aversion when PSD exclusively preceded the LS. For that purpose, we applied PSDs of 7.5, 10 and 20 hr duration respectively, preceding the session where rats were allowed to drink LiCl solution.

RESULTS

Figure 4 shows the reduction of learned avoidance under these particular conditions. It is worthwhile noting that the first occurrence of a significant reduction of learned avoidance as well as its evolution in relationship to the magnitude of PSD are not very different from the case (Experiment 2) in which post-LS PSD was systematically added after the pre-LS PSD. Thus only 7.5 hr of PSD were required to significantly ($p < 0.01$) increase the intake of NaCl solution compared to controls (Group C_2).

DISCUSSION

The experiment reported demonstrate that a learned taste aversion is completely abolished when the LS is preceded and followed by 96 hr PSD through the water tank technique. Shorter PSDs were less effective (Fig. 3). However, before the effects on learning can be attributed to PSD *per se* it is necessary to consider the shortcomings of this commonly used technique for PSD.

Firstly, placing the animals on a small pedestal above water clearly introduces a general stress component in addition to prevention of PS. Accordingly adrenal hypertrophy and increase of ACTH synthesis with thymus atrophy [10] have been reported under these conditions. Secondly, slow wave sleep is affected [3,11]. Unfortunately, alternative techniques for PSD are not readily available and it is theoretically impossible to conceive of a

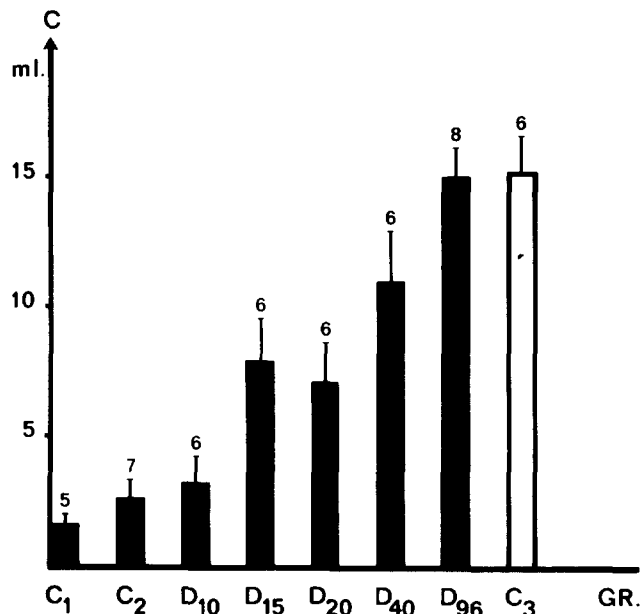


FIG. 3. Mean consumption and standard error (topped by N) of saline made aversive (except for C_3), by LiCl poisoning preceded and followed by increasing magnitude of PSD (D_{10} , D_{15} , D_{20} , D_{40} and D_{96}). Control groups (C_1 and C_2) were not deprived from paradoxical sleep.

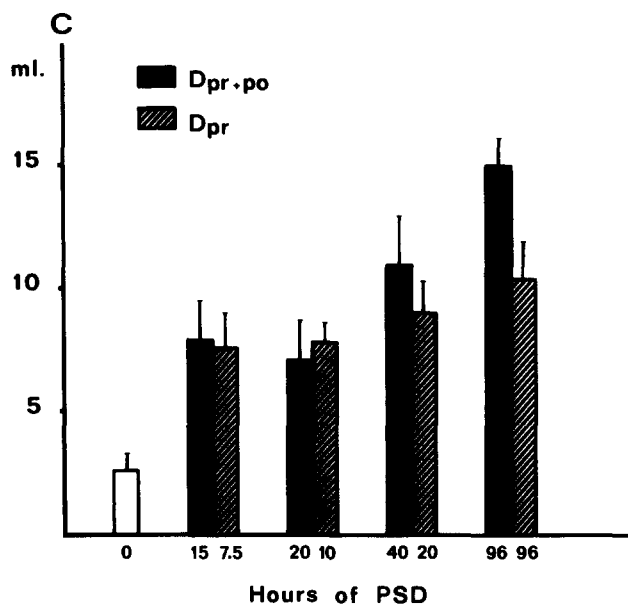


FIG. 4. Mean consumption and standard error, of saline when an increasing number of hours of PSD was applied only before (D_{pr}) the learning session, and when the same duration of PSD was added after the learning session ($D_{pr} + po$).

nonstressing PSD. We therefore employed a less than adequate control situation, namely placing animals on 20 cm pedestals (which preserve physiological sleep better than any smaller diameter pedestal [11]) which would also lead to nonspecific stressful aspects. These would include suspension above water, restricted movement during food and water deprivation, continuous illumination. It is noticeable that despite these nonspecific stresses these controls

(C₂) manifested the same degree of aversion learning as C₁ controls which had been maintained in their home cages. Whatever the contribution of nonspecific stress in affecting taste aversion learning, in the following discussion, when we refer to PSD, we emphasize the necessity to consider this is an ensemble of PSD and other factors.

Compared to classical or instrumental types of learning which had used the same technique, taste aversion learning is much more persistent through PSD. Several other attempts were made to abolish learned aversions using anesthesia [19] or electroconvulsive shock [15] and other manipulations. These manipulations applied between CS and US did not succeed in extinguishing taste aversion. This once again confirm that this type of learning must be distinguished from classical paradigms.

The crucial question posed in the light of Experiment 1 was whether these animals did not avoid the aversive solution because they could not acquire the avoidance or because they could not retain it, since PSD in that experiment both preceded and followed the LS. PSDs applied either before or after the LS (Experiment 3) indicated that pre-LS PSD was the principal factor affecting the learned aversion. It is reasonable to attribute this impairment to a lack of acquisition. In the same way, PSD following the LS would interfere with retention processes. Such an assumption is only indicative since it is difficult to assess that "acquisition" and "retention" processes can be so readily separated experimentally.

The suspicion that pre-LS PSD could introduce a "state dependant" learning of aversion can be ruled out by a number of relevant experimental studies. Using taste aversion learning, Burešová and Bureš [2] found that unilateral cortical spreading depression (CSD) does not interfere with learning despite the fact that this relevant

state occurs at LS. Also anaesthetics [19] and electroconvulsive shock [15] do not lead to state dependency of learning.

The fact that pre-LS PSD impairs taste aversion learning is in contrast to what is reported in classical learning paradigms where retention processes seem to be principally affected by PSD [8,9]. In fact, in prolonged PSD which tended to completely abolish taste aversion learning (Experiment 1) disturbance of retention appeared to play an additive role with the disturbance of acquisition (Fig. 2). However, for shorter periods of PSD there was no significant difference when the effect of post-LS was added to that of pre-LS PSD (Fig. 4).

It is possible that LiCl disrupts sleep patterns so as to abolish PS and thereby extends pre into post-LS PSD. However, the minimum PSD duration which was sufficient to significantly affect taste aversion learning was about 7.5 hr (applied before the LS). The addition of an equal duration of PSD post LS did not further affect animal's learning (Fig. 4). In addition, we have recently explored the effect of LiCl on electrophysiological correlates of sleep. Our data clearly show that during the 7.5 and 24 hr after LiCl treatment there is no disruption of PS, and in fact there is an increase (Danguir, unpublished data).

In conclusion, it appears that learned taste aversion, besides its wellknown particularities is a tough and persistent process. PSD of durations known to completely abolish classical learning leave the taste aversion virtually unaffected. Whatever the nature of the central (neurochemical) changes involved in taste aversion learning, those components implicated in the acquisition processes appear to be more severely disrupted by PSD (and attendant nonspecific stress) than are those of retention.

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